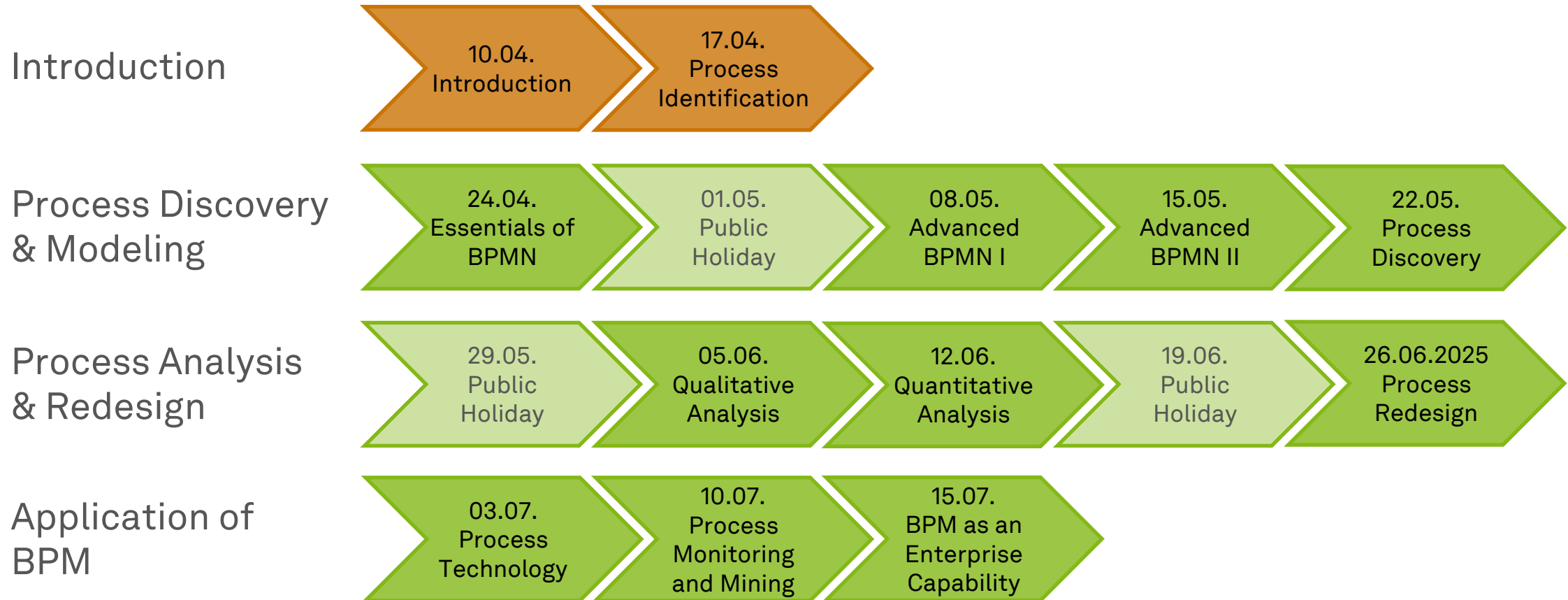


Business Process Management

Essentials of Process Modeling



Roadmap



Learning Goals of Today

- Understand the purpose of process modeling
- Understand the purpose of BPMN
- Get to know BPMN's basic modeling constructs
- Be able to use basic BPMN

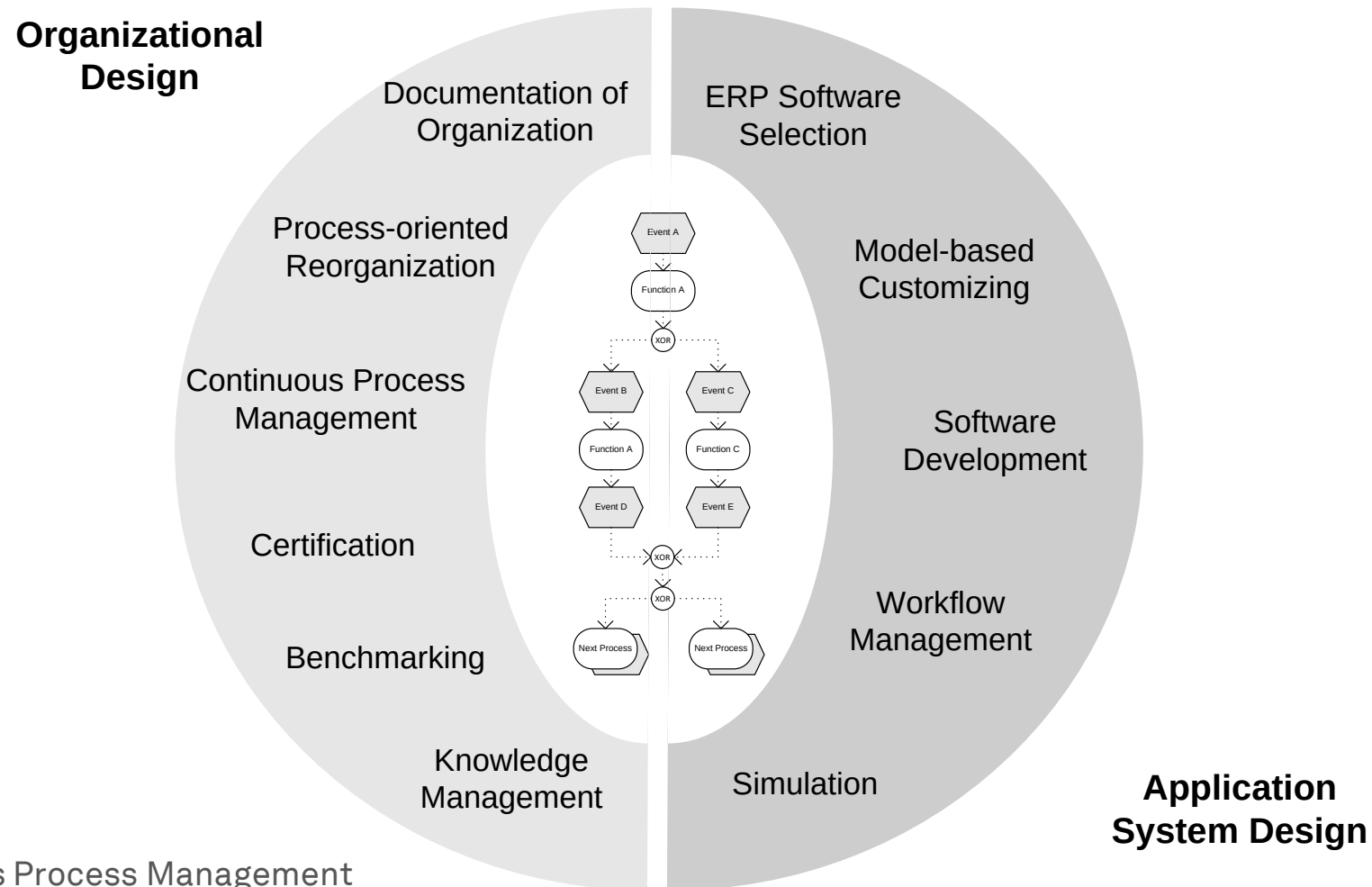
Purpose of Process Modeling



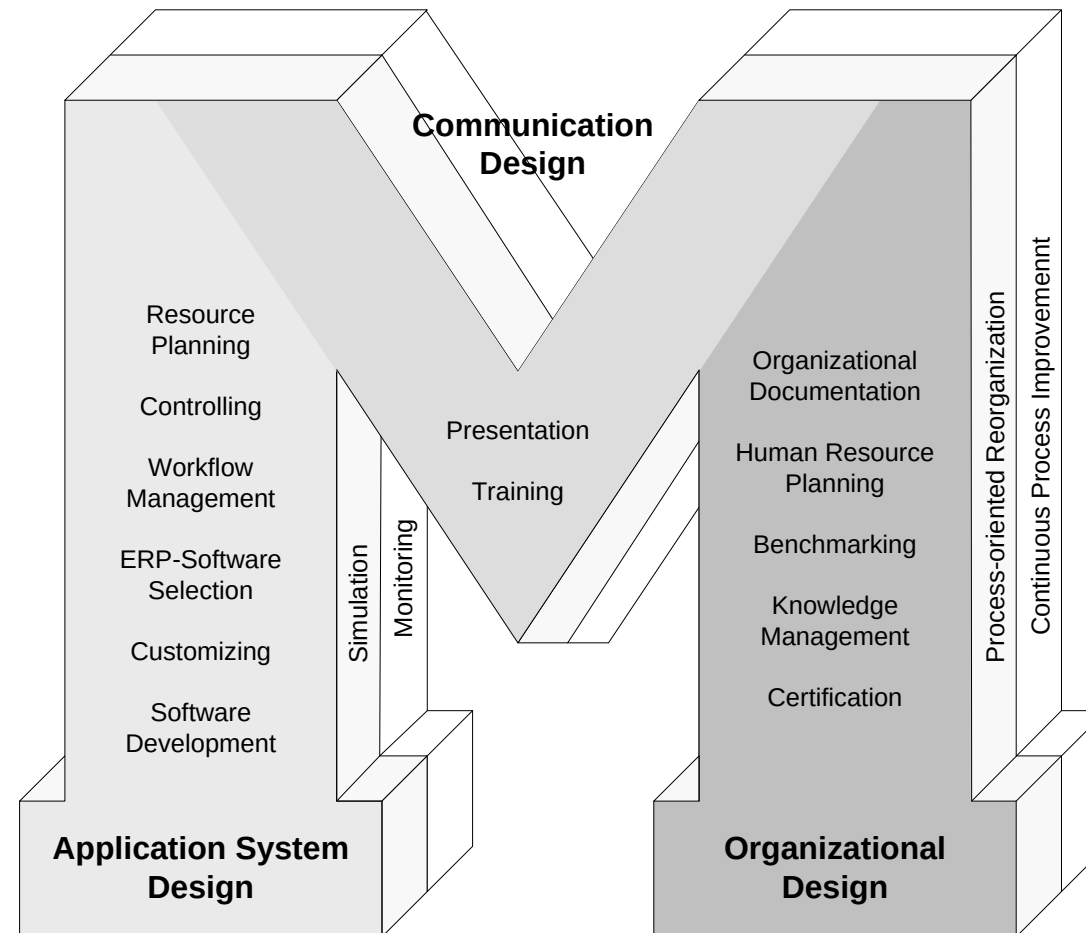
Why Process Modeling?

- **Understand** of the process with the people who are involved with it
- **Common language** for naming and framing an issue
- Provides a blueprint for **process automation**
- **Integrates** processes with other artefacts
 - e.g., systems, organizations, data, etc.
- Enables **walk-through**, **validation**, and **testing**
- Use as a **benchmark** for measuring improvements

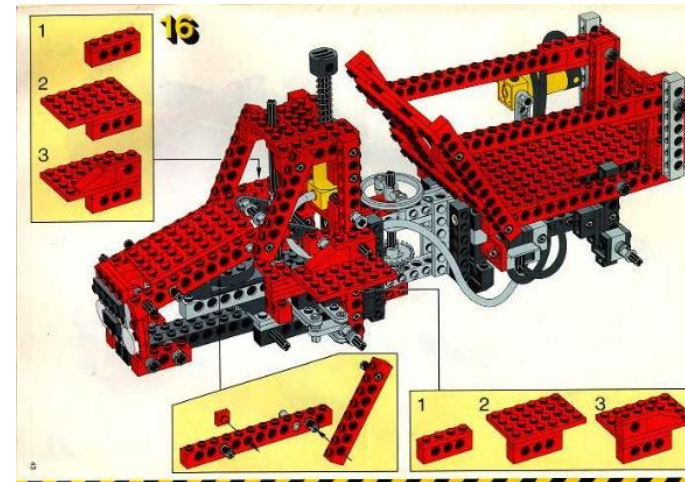
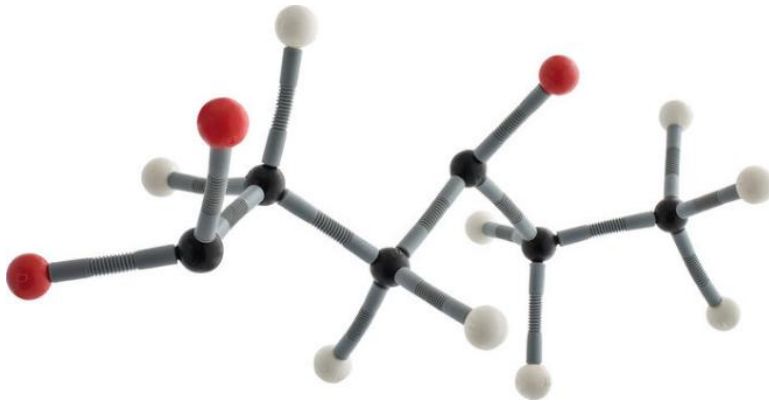
Purposes of Process Modeling (1)



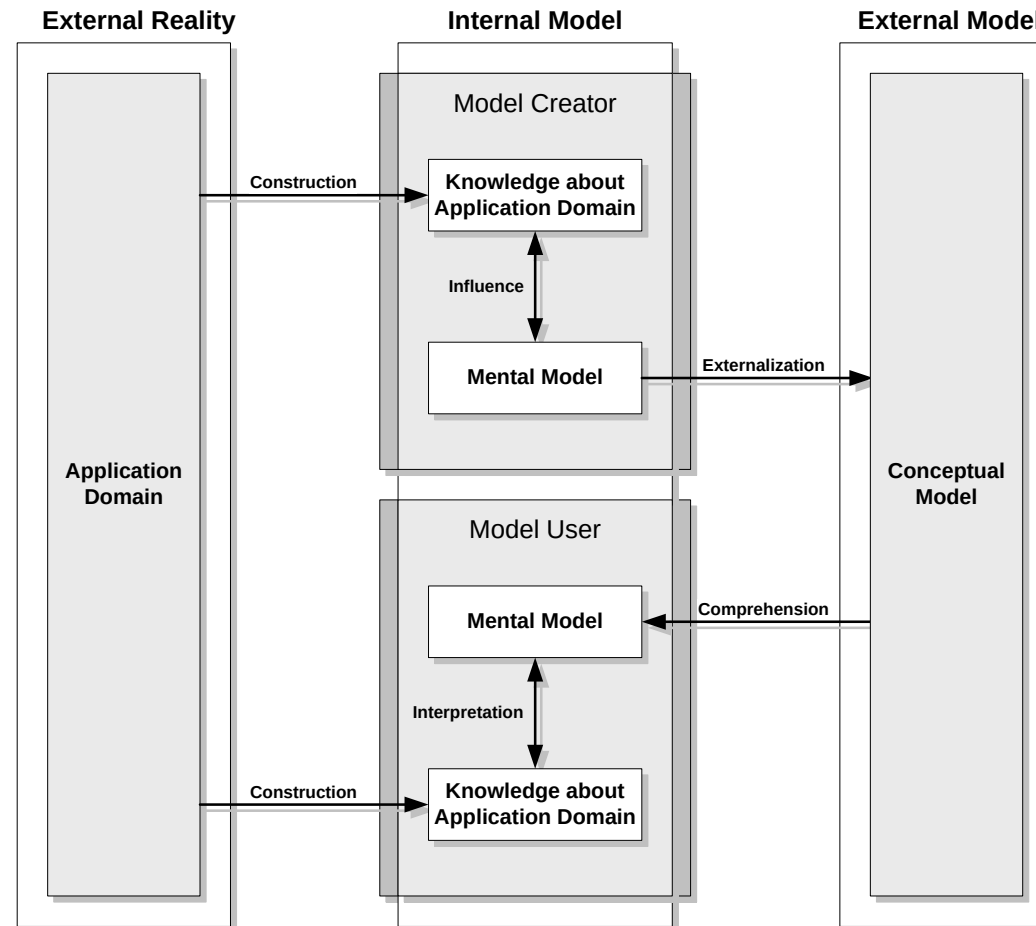
Purposes of Process Models (2)



What Actually is a Model?

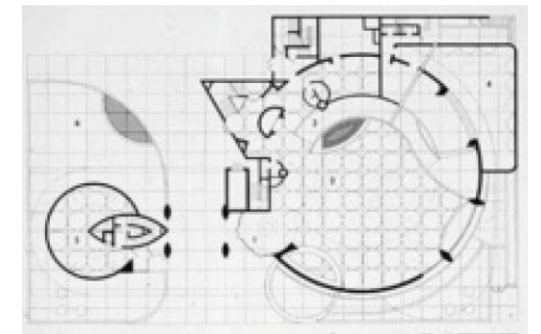


Mental Model



Features of a Model?

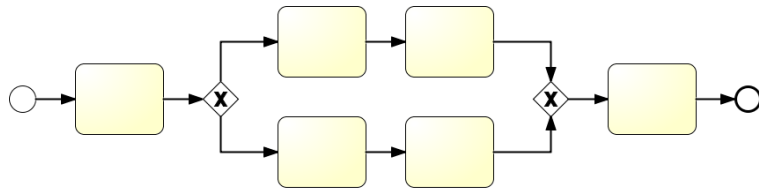
- **Mapping**
 - Models are depictions or **representation** of a natural or an artificial original, which themselves can be models again
- **Abstraction**
 - Models are **abstractions** from real world phenomena
 - Models are developed for the purpose of **reducing** overall **complexity**
- **Pragmatic purpose**
 - Models may not be clearly associated to its appropriate original
 - Models serve as a **substitute**
 - a) for certain subjects
 - b) within certain time intervals
 - c) under constraints imposed by certain operations



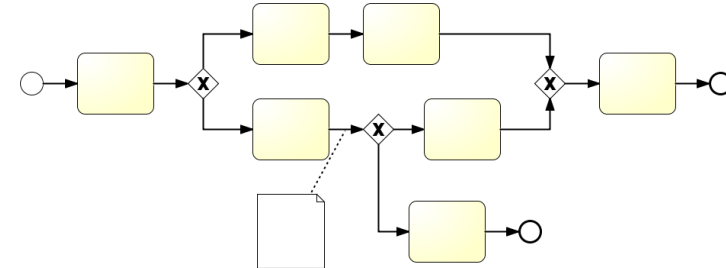
Stachowiak (1973), Dumas et al. (2018)

Objectives of Modeling

- Descriptive Modeling
(as-is-processes)

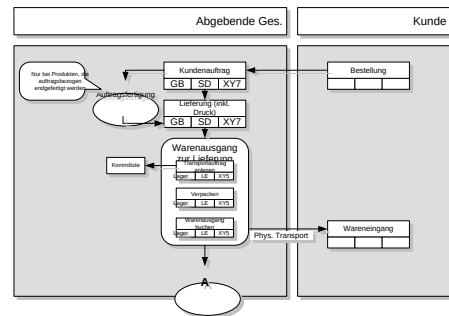


- Prescriptive Modeling
(to-be-processes)

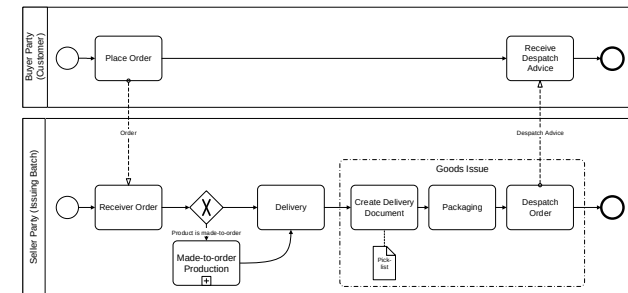


Business Process Modeling

- Business processes models are being developed in the purpose...
 - of a specific modeling **subject**,
 - for a specific target **audience**
 - with a specific modeling **purpose** in mind



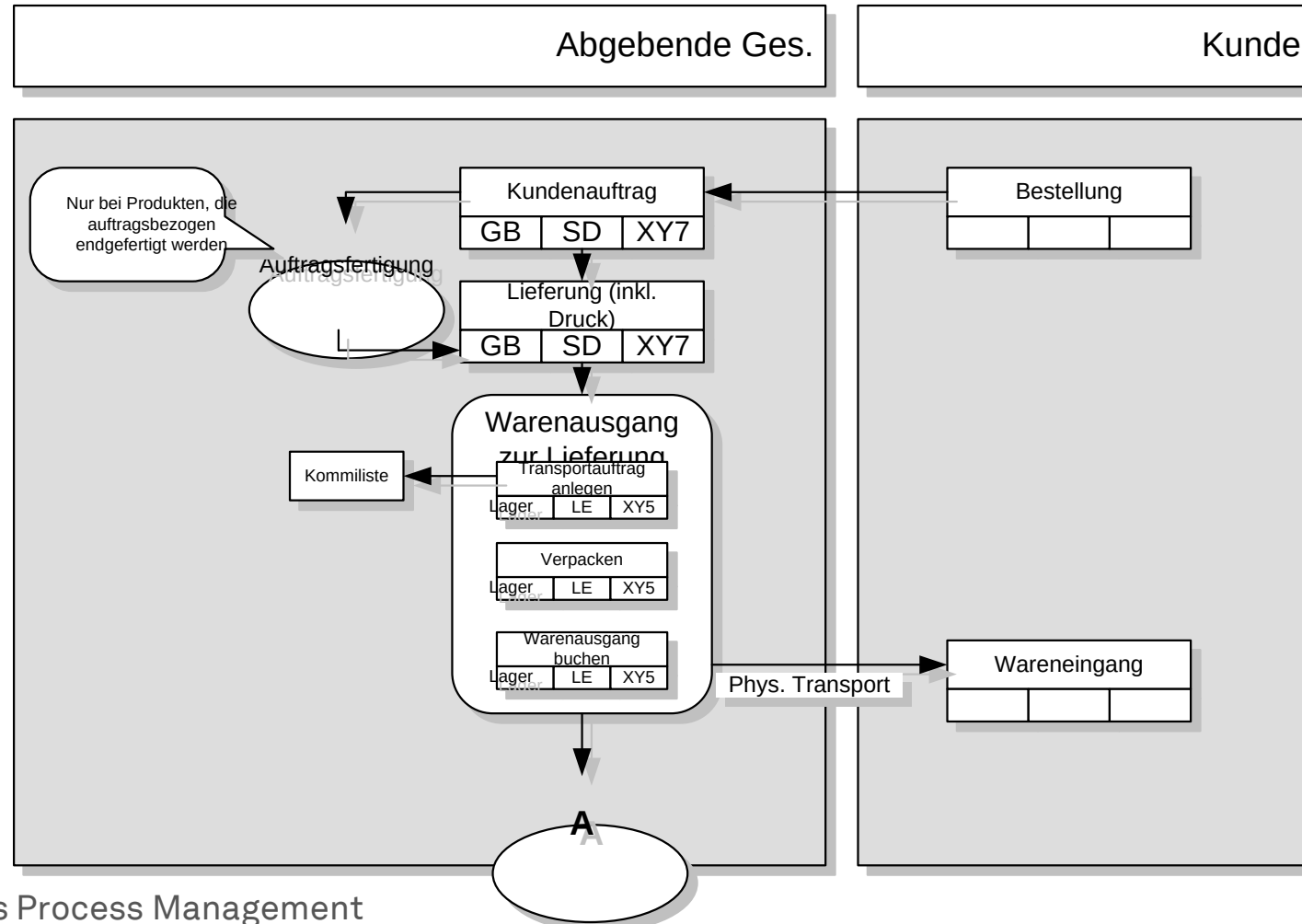
confusing & complex model



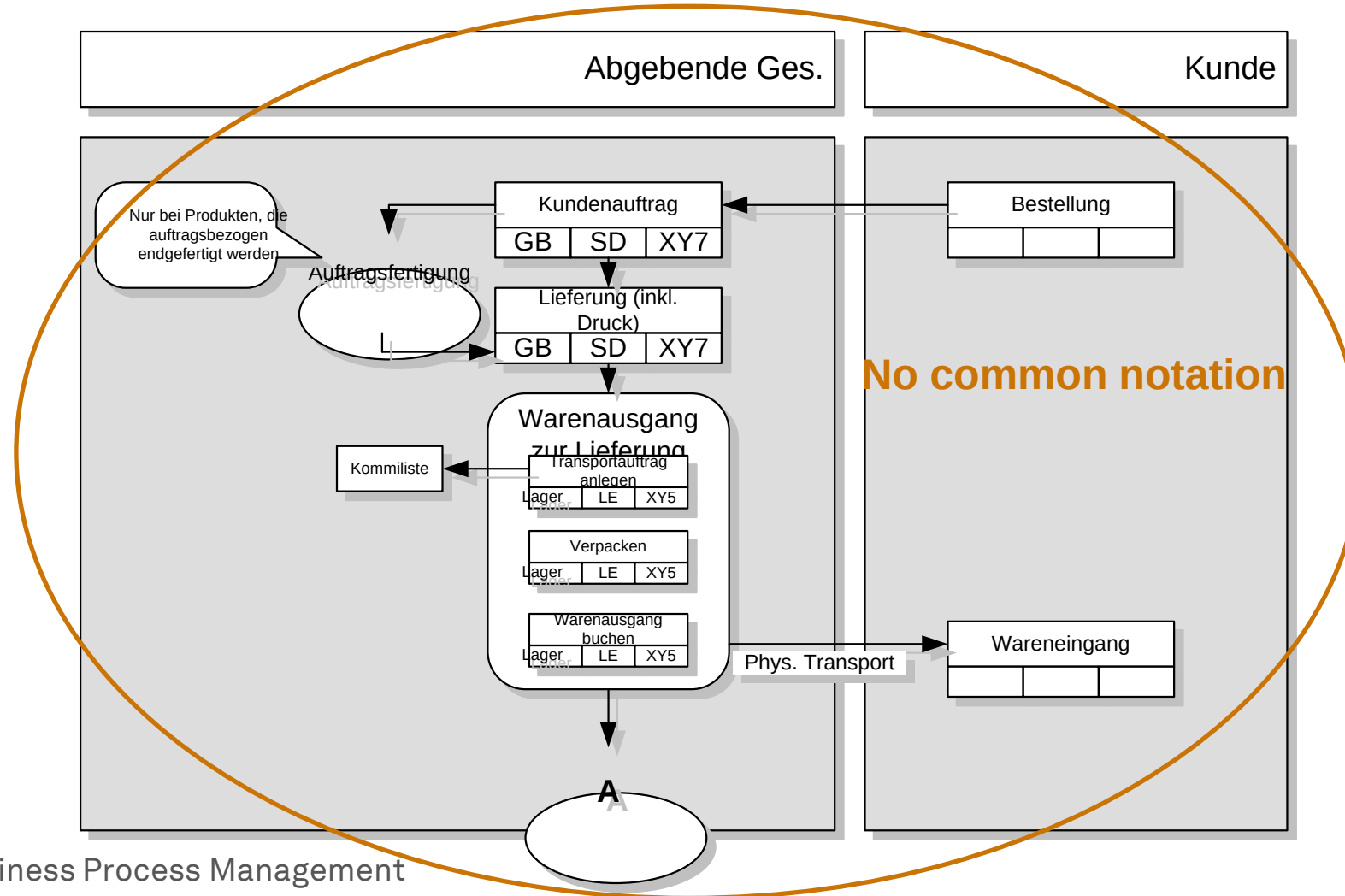
clear and easy to understand

- There is no **right** or **wrong**, but there is **relevant** and **irrelevant**

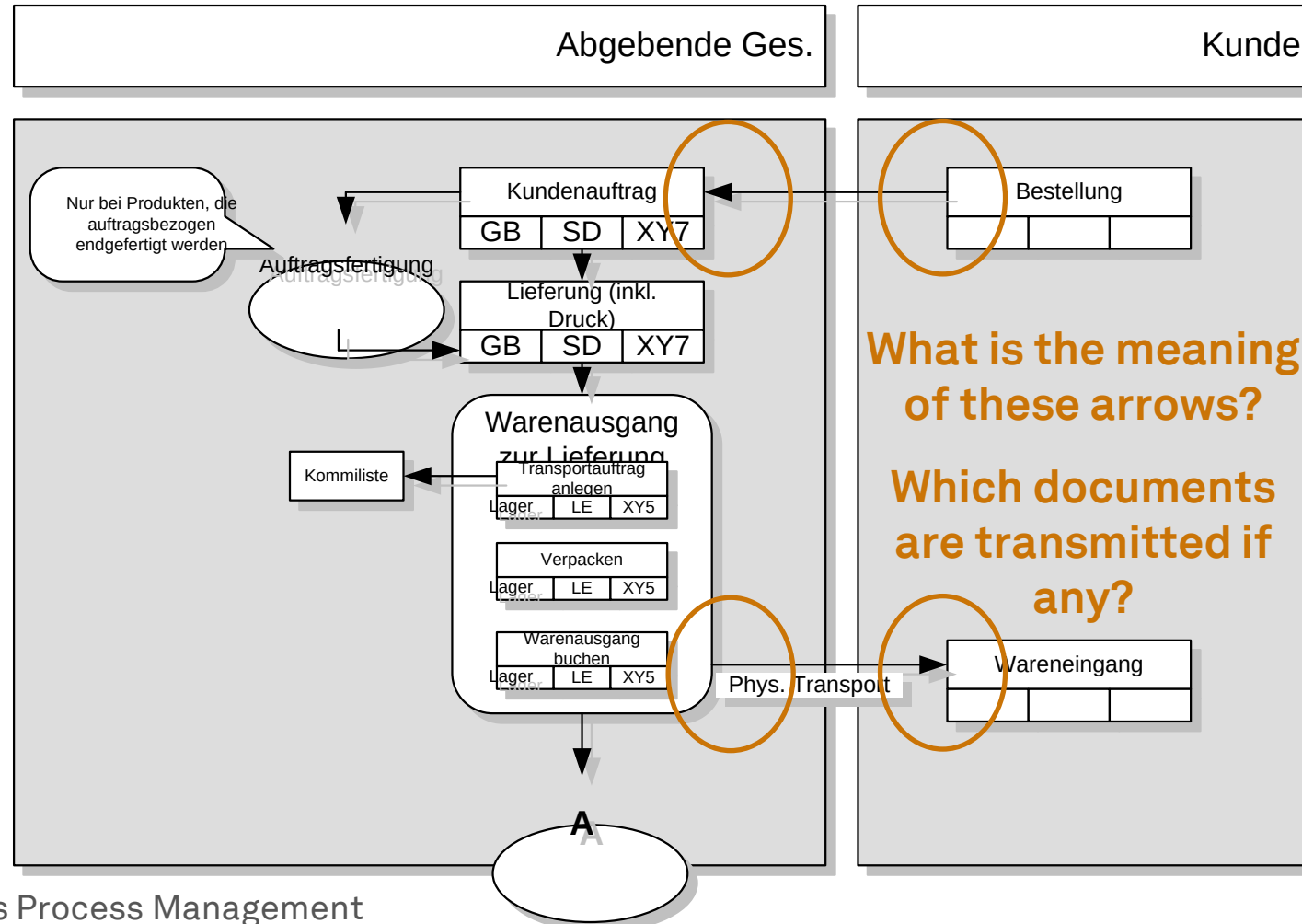
Problems with Process Documentation



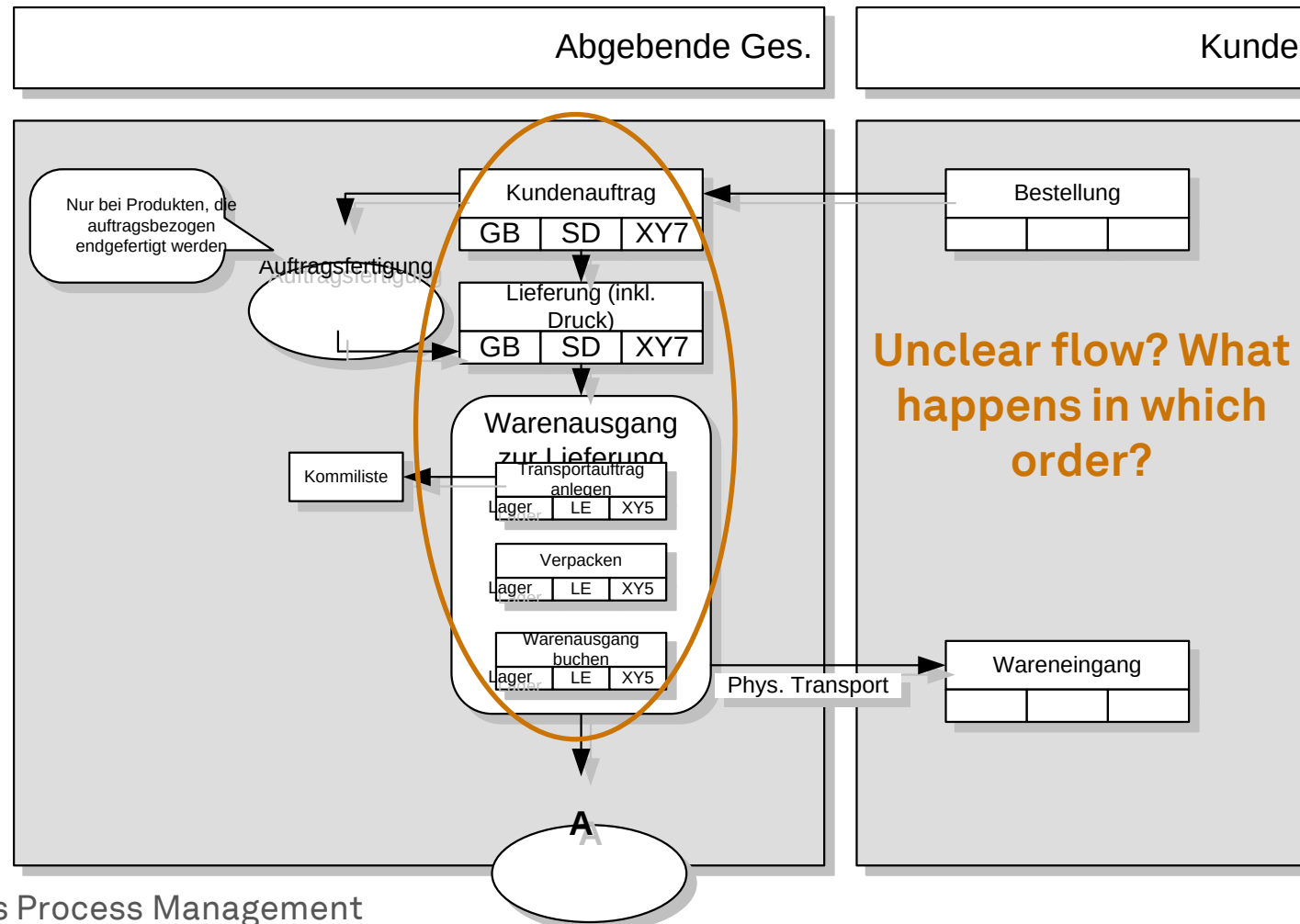
Problems with Process Documentation



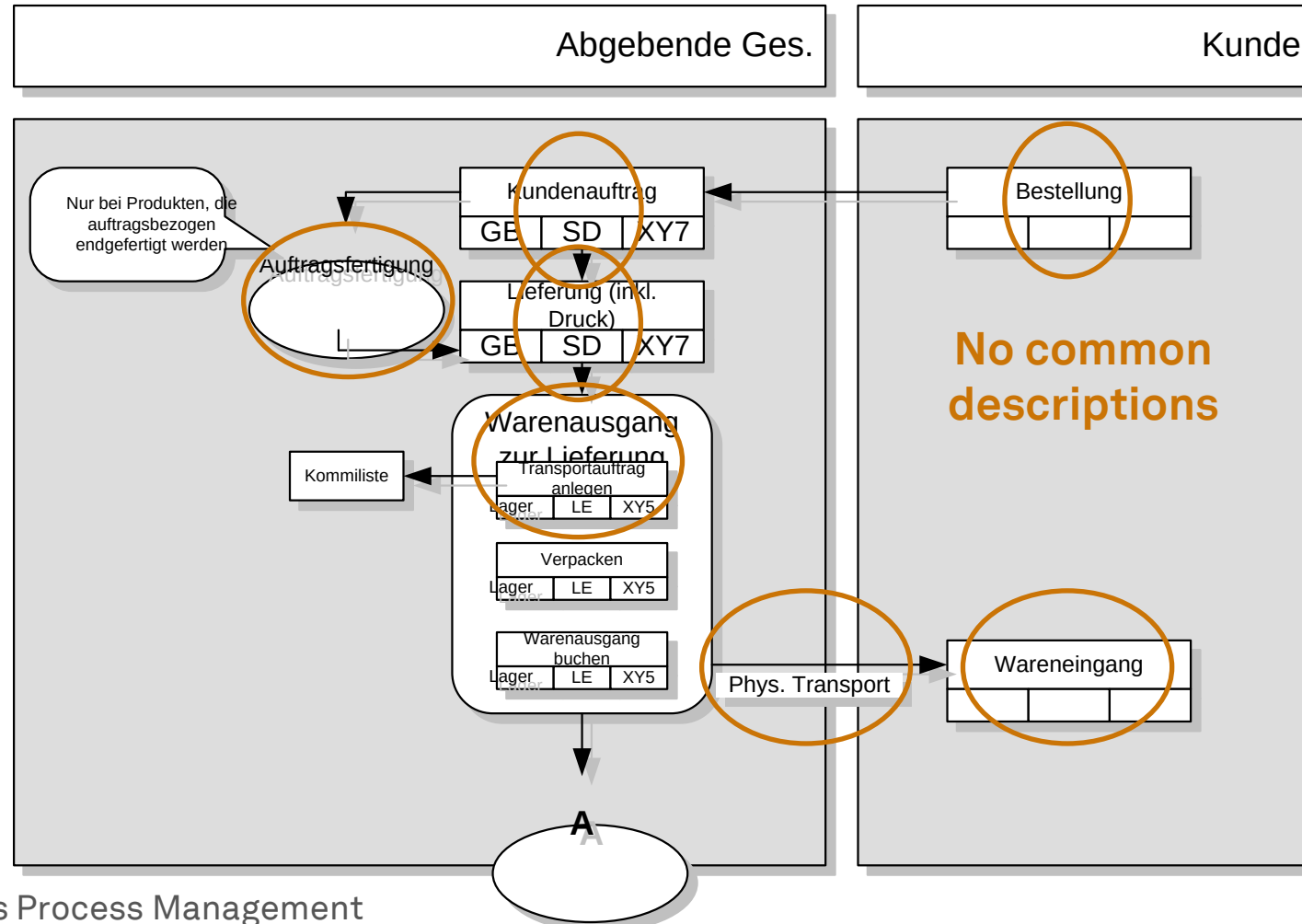
Problems with Process Documentation



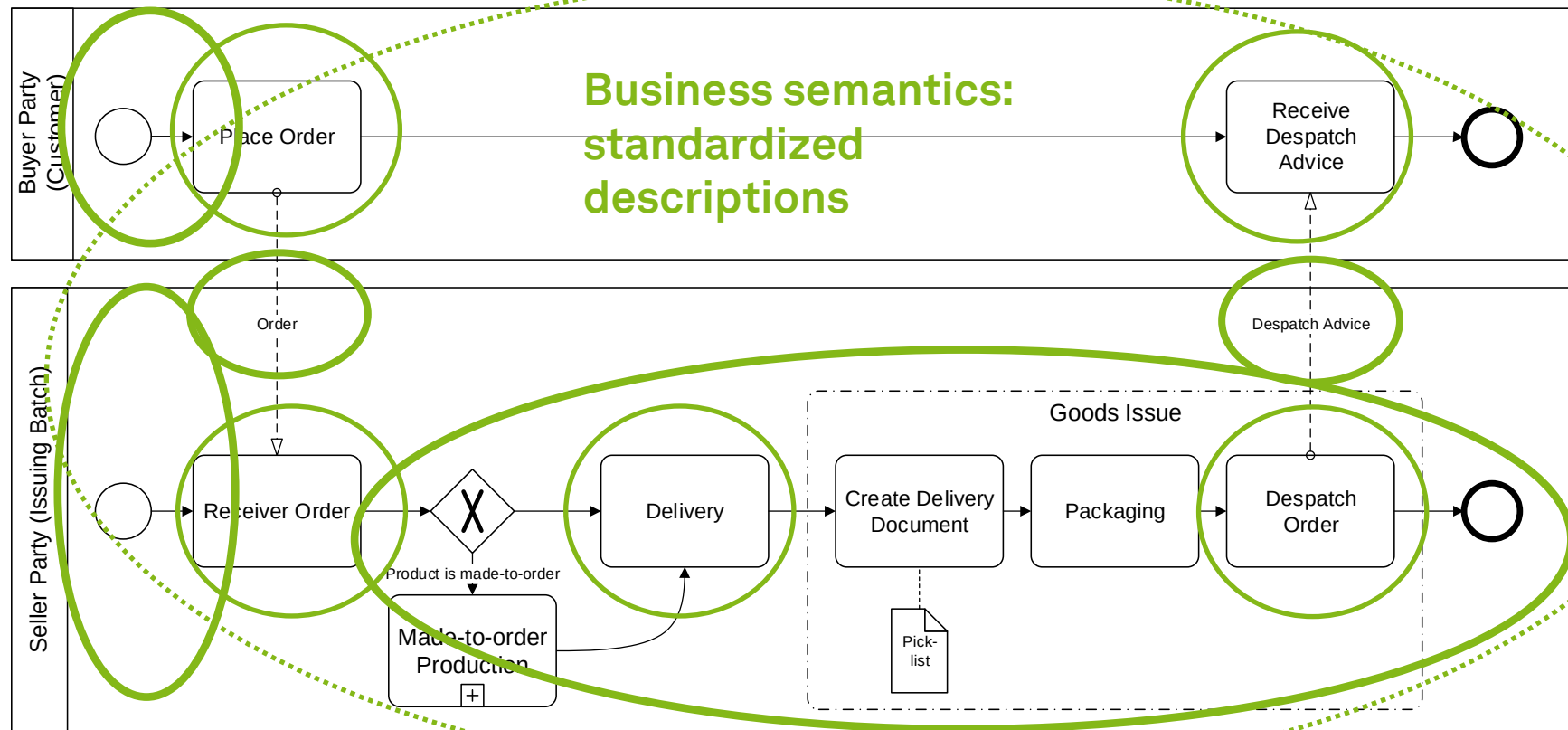
Problems with Process Documentation



Problems with Process Documentation

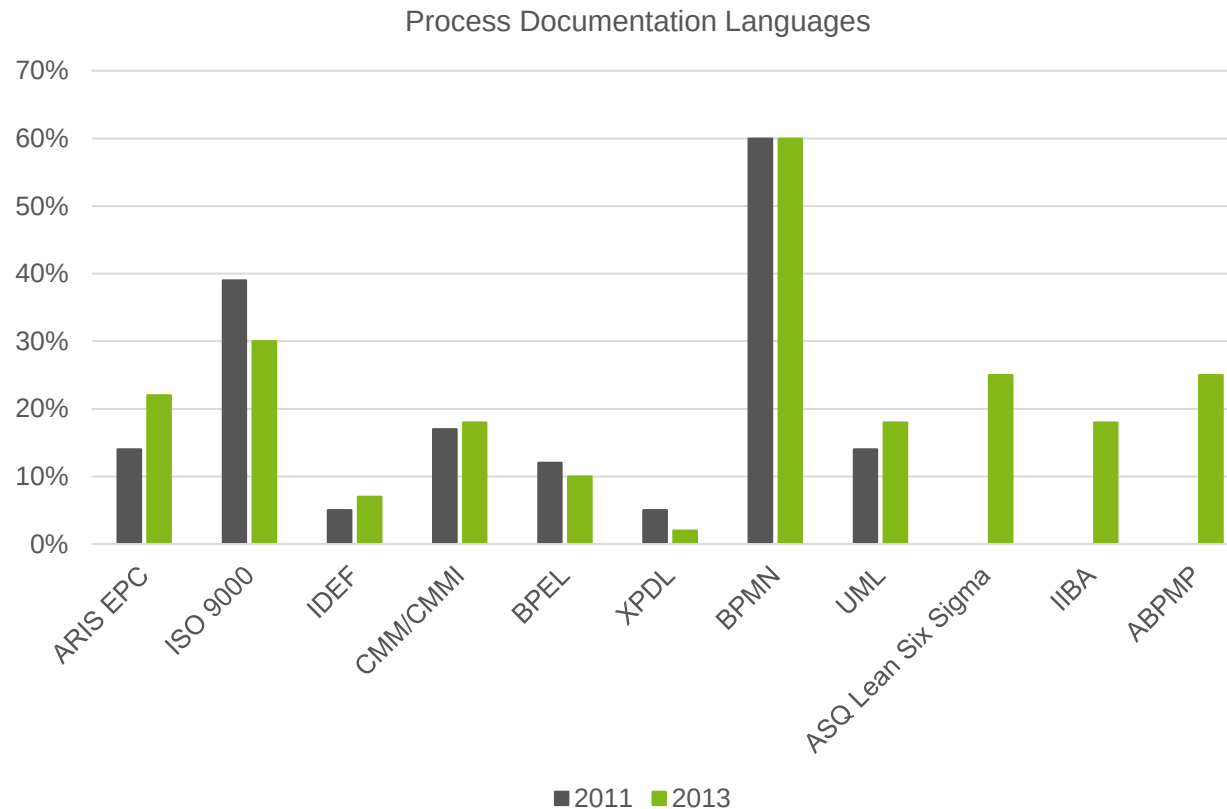


Standardized Process Documentation with BPMN



There are many different notations for process modeling...

- ...but BPMN is the (de facto) standard



Derived from BPTrends (2014)

Business Process Model and Notation (BPMN)



Origins of BPMN

- The Business Process Management Institute (BPMI) – now merged into the OMG – developed the Business Process Modeling Language (BPML) and **realizes the need for a graphical representation**
- In 2001, the BPM Notation Working Group was formed
- Business Process Modeling Notation (BPMN 1.0/1.1/1.2)
 - May 2004: First BPMN specification was released to public
 - February 2006: BPMN specification was adopted by OMG
 - February 2008: BPMN 1.1 was released to public
 - February 2009: BPMN 1.2 was released to public

Origins of BPMN

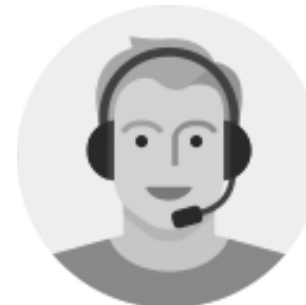
- Business Process **Model** and **Notation** (2.0/2.01/2.02)
 - January 2011: **BPMN 2.0** was released to public
 - Name changed due to changing the scope from “pure” graphical representation to a process definition language
 - Major update, focus on execution
 - V2.01 also published as ISO/IEC 19510

Design Goals for BPMN (1.x)

- Must be acceptable and usable by the **business community**
- Must be able to generate **executable** processes (e.g., XPD, BPEL) through a BPMN Model
- Should **bridge the gap** between business analysts and IT-experts
- BPMN is intended to be **methodology agnostic**
 - BPM methodologies will give guidance as to the purpose and level of detail for modeling
 - BPMN is as complex as it needs to be

BPMN 2.0 Scope

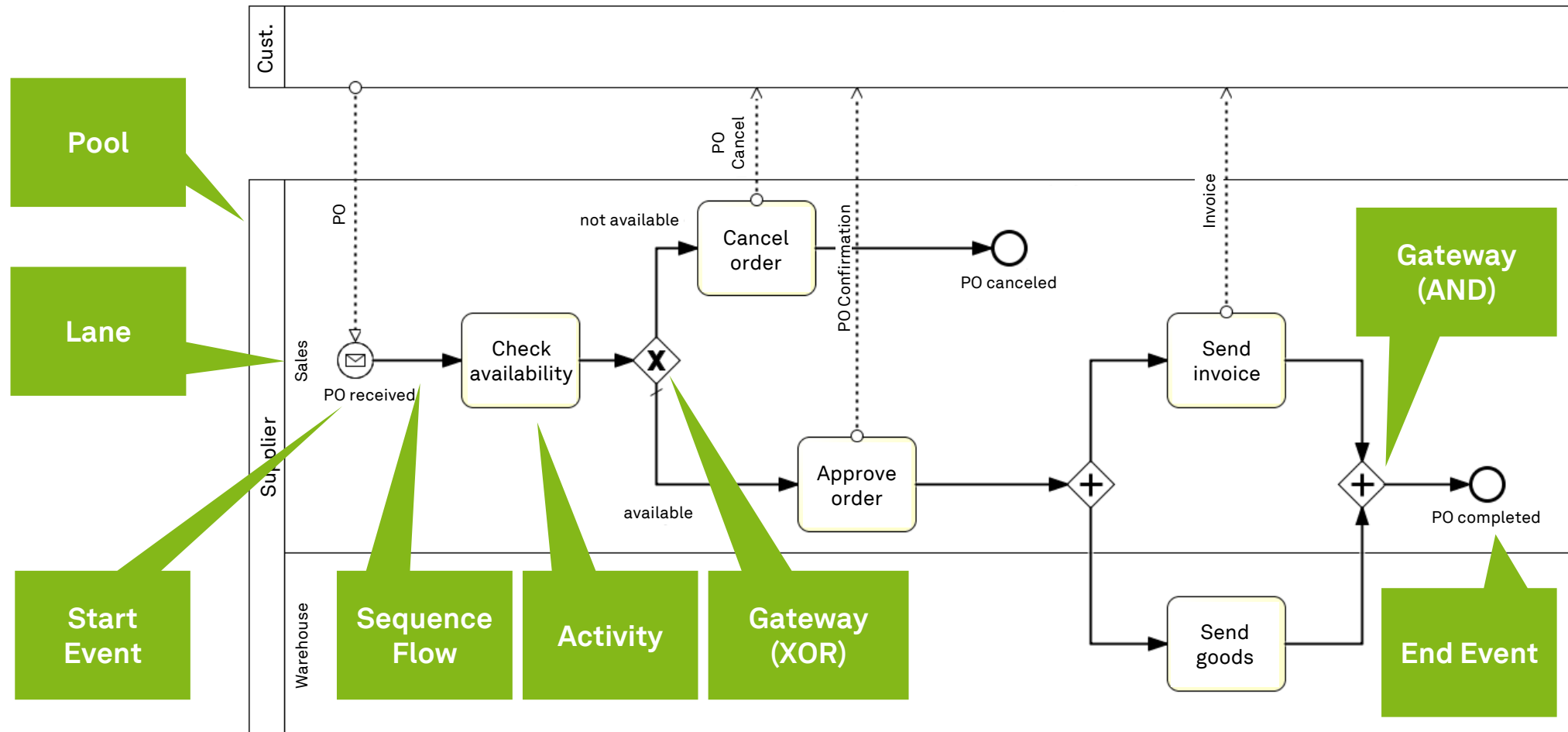
- The **primary goal** of BPMN is to provide a notation that is
 - readily **understandable** by **all business users**,
 - from the business analysts that create the initial drafts of the processes,
 - to the technical developers responsible for implementing the technology that will perform those processes,
 - and finally, to the business people who will manage and monitor those processes



BPMN 2.0 Objects and Diagram Types

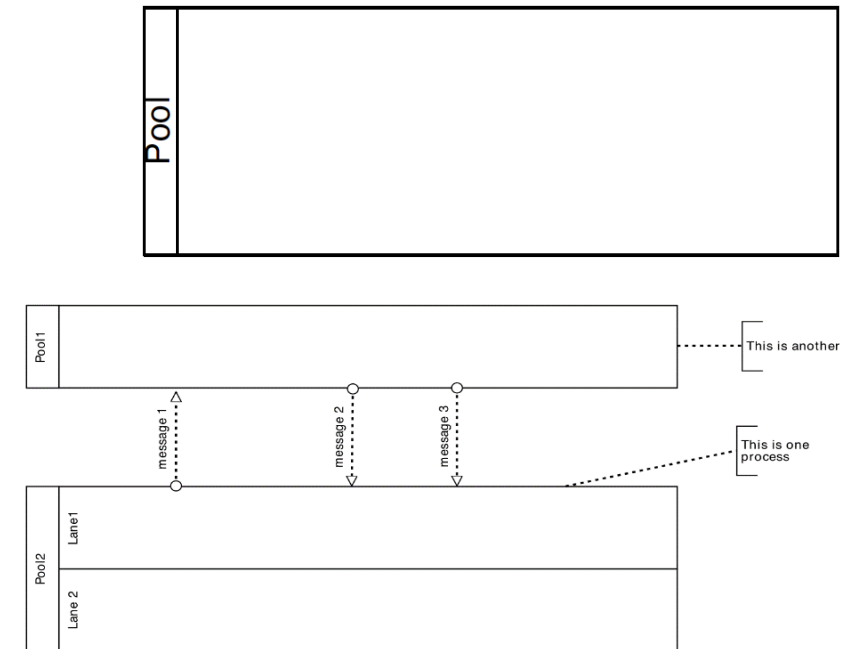
- BPMN specifies a **combination of objects**, their **graphical representation** and **supporting attributes** (preliminaries for implementation and enactment)
- Diagram Types
 - **Collaboration Diagram**
 - Choreography Diagram
 - Conversation Diagram

A Simple Process Model with BPMN 2.0



Structuring Elements in Process Diagrams: Pools

- A **Pool** partitions a **set of Activities** from other Pools
- A Pool is a graphical representation of an active **resource**
- Represents **independent organizational entities**
 - Customer
 - Supplier
 - etc.
- It can be a **participant, software, or hardware**
 - e.g., an ERP system
 - e.g., a manufacturing plant



Structuring Elements in Process Diagrams: Lanes

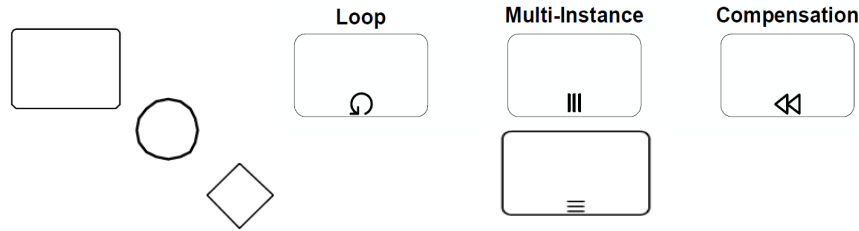
- A **Lane** is a sub-partition within a Pool
- Lanes **extend** the **entire length** of a Pool either horizontally or vertically
- Lanes can be **nested**
- Lanes are used to **organize and categorize** Activities (typically by **roles**)
- Represent resource classes in the same organizational unit or space
 - e.g., sales department, customer care, logistics, manager, etc.



Overview of Basic BPMN Objects in Process Diagrams

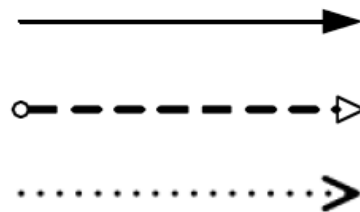
- **Flow Objects**

- Activity
- Event
- Gateway



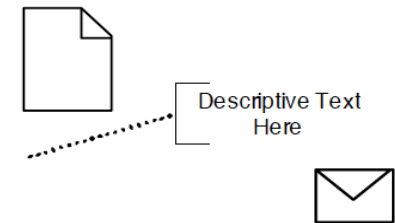
- **Connectors**

- Sequence Flow
- Message Flow
- Association



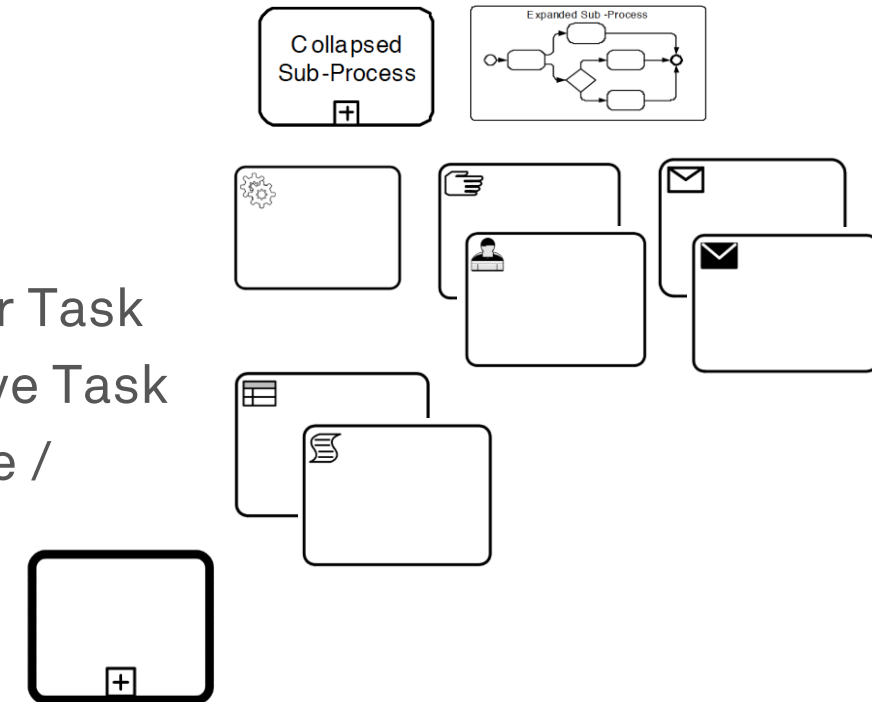
- **Artifacts**

- Data Object
- Text Annotation
- Message



Activities and Tasks in BPMN

- Activities represent **units of work** that have a duration **performed** within a business process
 - Executable elements of BPMN process
 - Can be **atomic** (Task) or **compound** (Sub-Process)
 - Carry attributes such as **loopCharacteristics** or **maximumLoop**
- Types of Activities in a process
 - Sub-Process
 - Task
 - **Service Task**
 - Manual / User Task
 - Send / Receive Task
 - Business Rule / Script Task
 - Call Activity



Events in BPMN

- An event is **something** that happens **instantaneously** during the course of a process
- Events **affect the flow** of a Process, usually have a cause or an impact and allow for reaction
- Types of Events
 - **Start** Events
 - **End** Events
 - **Intermediate** Events
- Types of Event behavior
 - Catch a trigger (message, timer, ...)
 - Throw a result (eventually caught by another event)



There are quite some events in BPMN

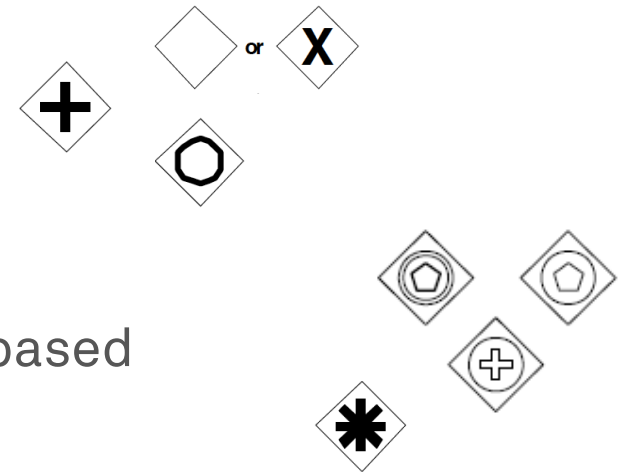
Types	Start			Intermediate				End
	Top-Level	Event Sub-Process Interrupting	Event Sub-Process Non-Interrupting	Catching	Boundary Interrupting	Boundary Non-Interrupting	Throwing	
None								
Message								
Timer								
Error								
Escalation								
Cancel								
Compensation								
Conditional								
Link								
Signal								
Terminate								
Multiple								
Parallel Multiple								

Gateways in BPMN

- Gateways are used to **control** how **Sequence Flows** interact as they converge and diverge within a Process
- Employ **gating mechanism** that either allows or disallows passage through the Gateway
- As Tokens arrive at a Gateway, they can be **merged** together on input and/or **split** apart on output

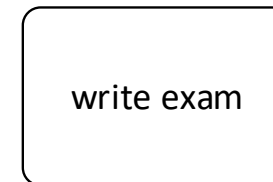
- Types of Gateways

- **Exclusive**
- **Parallel**
- **Inclusive**
- Event-based
- Parallel Event-based
- Complex

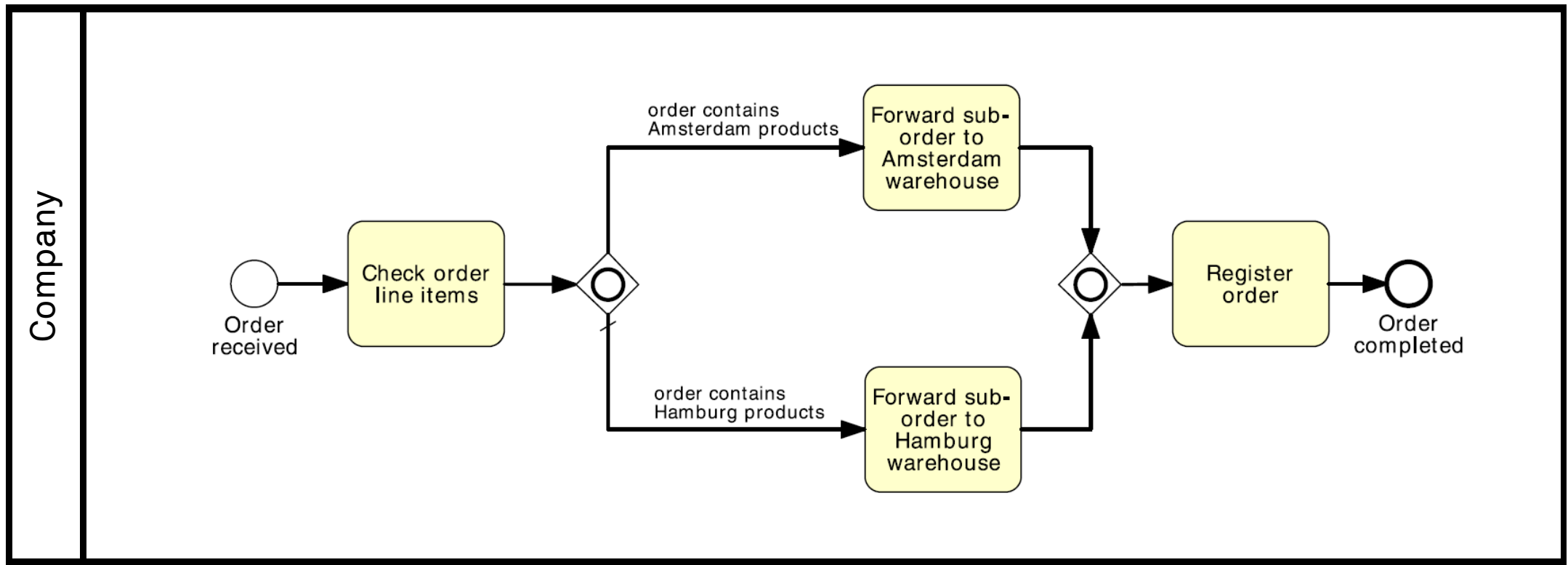


Naming Conventions

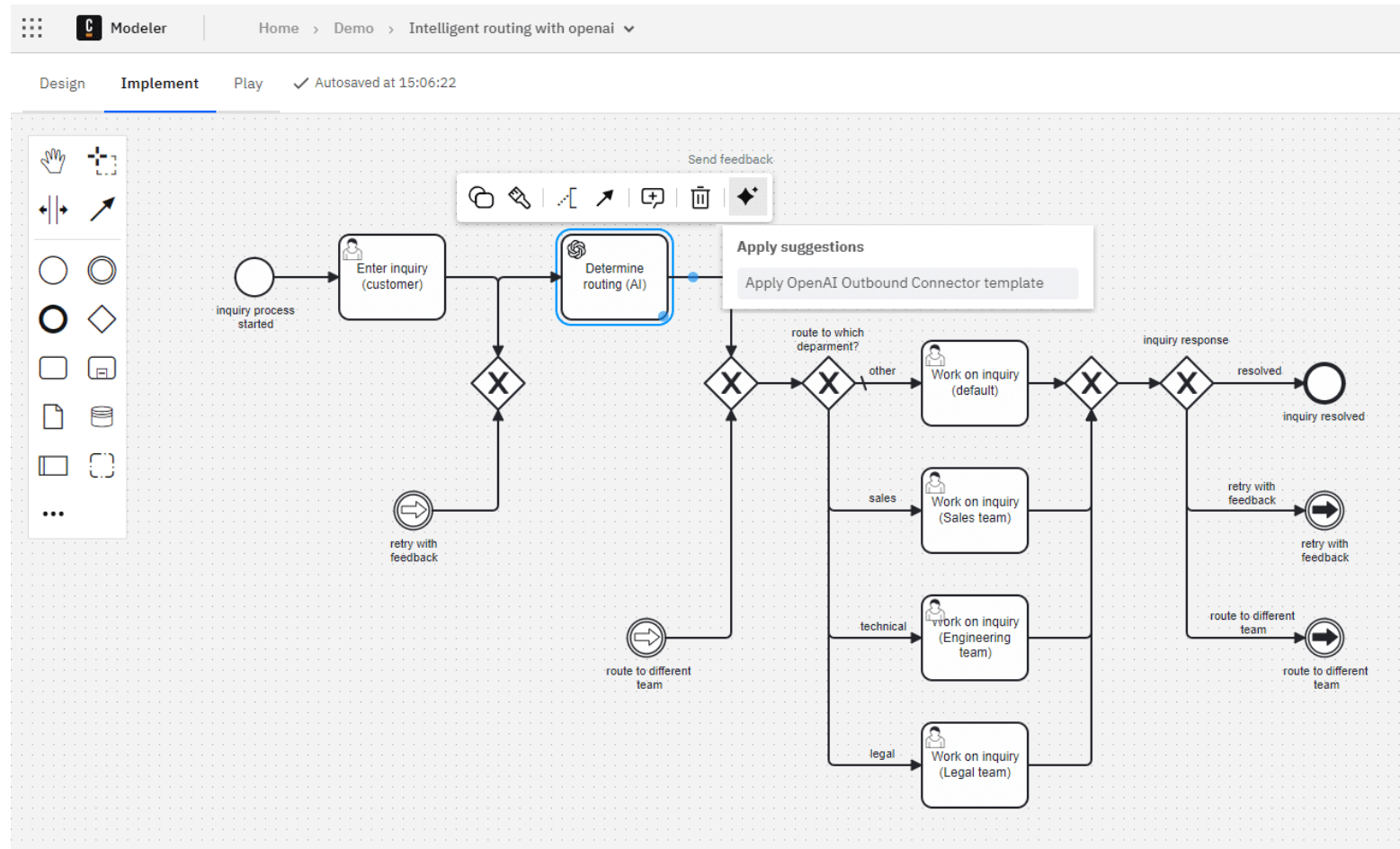
- There is **no formal restriction**, no overall naming convention
- It is **good practice** to name
 - Activities and tasks
 - **imperative verb followed by noun** (e.g., “write exam”)
 - Events
 - **Noun followed by past participle verb** (e.g., “exam time ended”)
 - Process model
 - **Nouns** (e.g., “exam marking” rather than “mark exams”)



A Simple BPMN Model

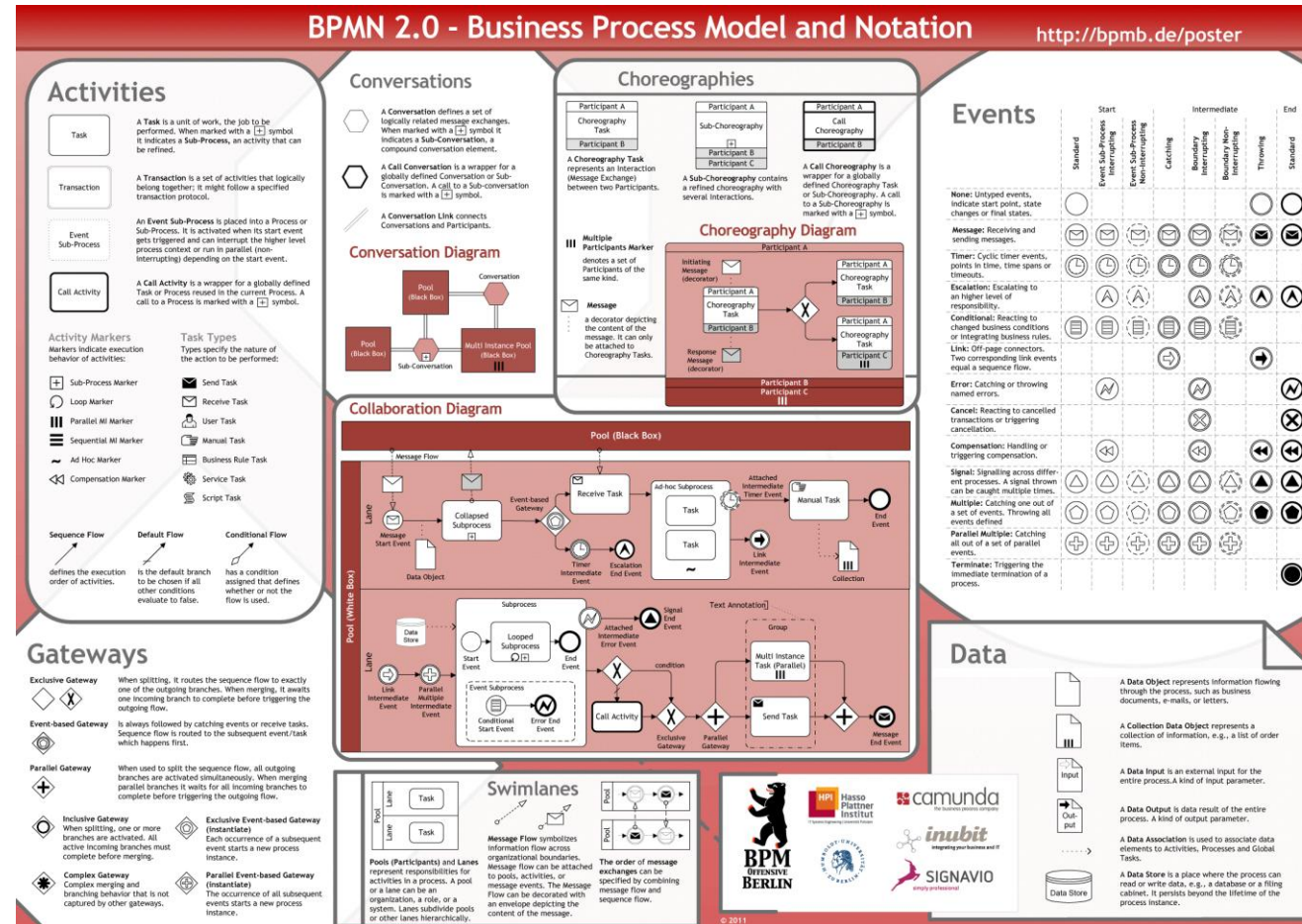


BPMN Tools - Camunda



<https://camunda.com/platform/modeler/>

BPMN 2.0 Poster



Learning Goals of Today

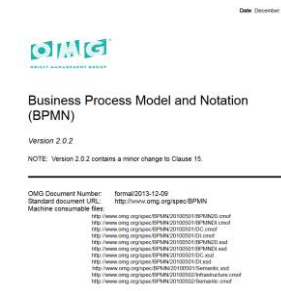
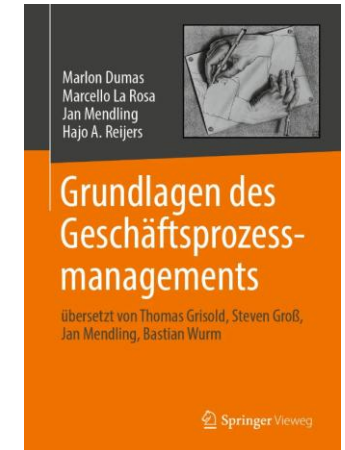
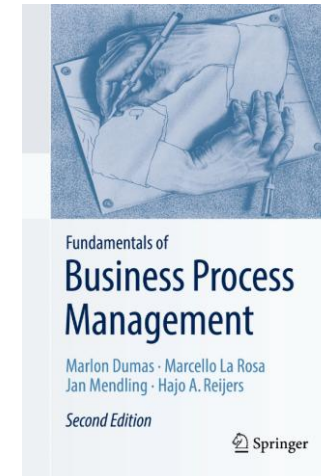
- Understand the purpose of process modeling
- Understand the purpose of BPMN
- Get to know BPMN's basic modeling constructs
- Be able to use basic BPMN

Recap Questions

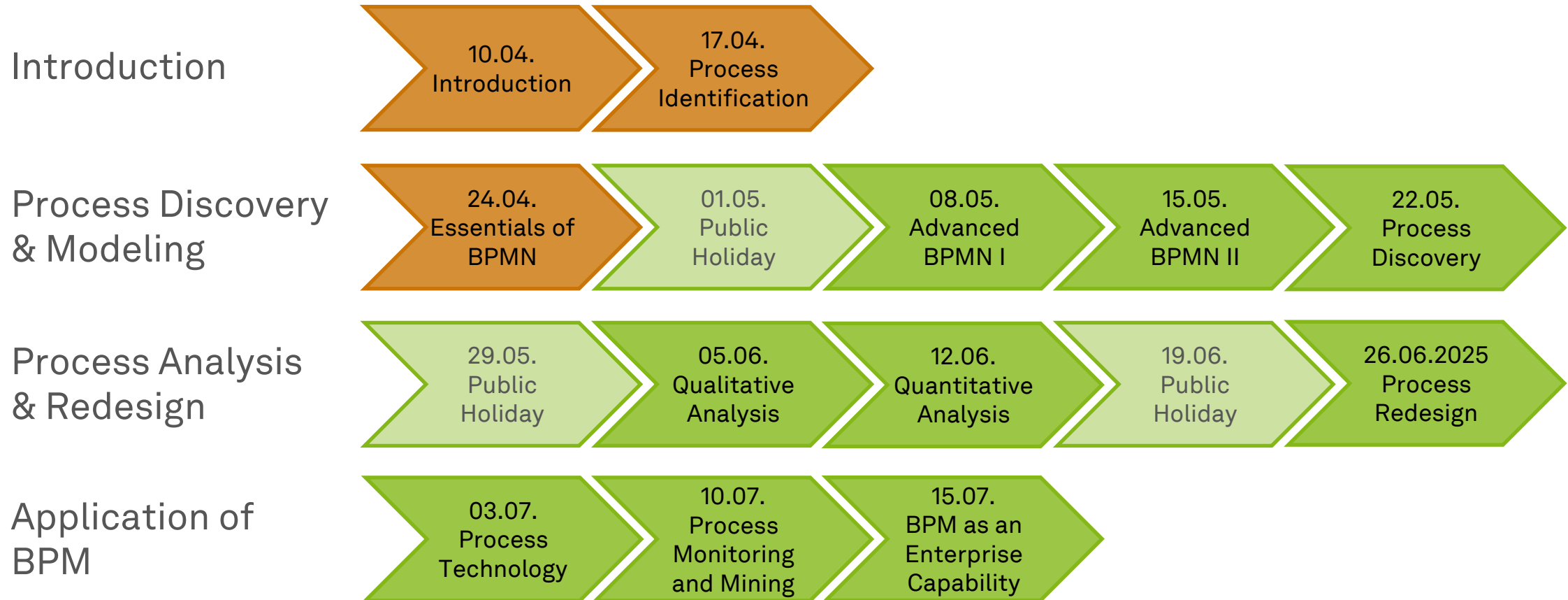
- What are features of a model and why do we model processes?
- What are pools and lanes in BPMN model?
- What is the difference between activities and events in a BPMN model?
- Why do we need gateways in a BPMN model?

Literature

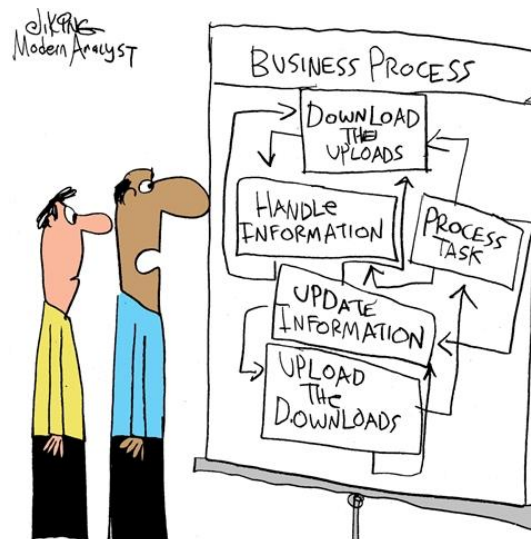
- **Section 3: Essential Process Modeling**
- **Business Process Model and Notation (BPMN)**
 - <https://www.omg.org/spec/BPMN/2.0.2>



Roadmap



Any Questions about Process Modeling?



*"Charlie, did you really get these details
from the customer?"*



ec.cs.tu-dortmund.de

